1 Requested-error theta samples for three fixed conductors

For the primitive characters $\chi_3 = (0, 1, -1)$, $\chi_4 = (0, 1, 0, -1)$, $\chi_5 = (0, 1, -1, -1, 1)$ with parities $a = 1, 1, 0$, retain the classical primitive theta/Mellin identity and root number one already stated above. Put $D = [-1, 2] + i[-4/3, 4/3]$ and $\delta = 3/q$. The following is the mathematical specification of an adaptive extension; implementation acceptance is a separate check.

Theorem 1 (Finite adaptive sample specification)**.** *For every supported conductor, exact rational $s \in D$, and rational $\eta > 0$, the choices below terminate and give a rational complex midpoint y and exact rational radius $R \le \eta$ with $|y - \Lambda_q(s)| \le R$. The same conclusion holds for a specified true circle node when its root and node enclosures are refined on every precision retry.*

Proof. For $x \ge 0$ define the exact rational bound $E(x) = 2^{-\lfloor x \rfloor}$. Since $e > 2$, $e^{-x} \le E(x)$; this bound is nonincreasing and tends to zero. Let $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$ and $C_0 = 1$, $C_1 = (1 + \delta^{-1})/2$. For $N \ge 1$, $n_0 = N + 1$, $J \ge 1$, $T = 2^J$, use

$$B_n = 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)},$$

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

Here $\delta n_0 \ge 6/5$ ensures a positive denominator; $E(x)$ equals one for $0 \le x < 1$, so this safeguard is essential. On the four panels of each $[2^j, 2^{j+1}]$, write midpoint m, halfwidth h, $\rho = m/2$, $v = h/\rho \le 2/9$, and $B(m) = \max\{(m + \rho)^2, (m - \rho)^{-2}, 1\}$. Use the Cauchy remainder

$$B_Q = \sum_{\text{panels}} \sum_{n=1}^{N} \frac{2n^a \, 2h \, 3B(m) E(\delta n^2 m/2) v^{2K+2}}{(2K + 3)(1 - v^2)}.$$

Choose in turn the first $N \ge 1$, first $J \ge 1$, and first $K \ge 0$ for which their respective budgets are at most $\eta/8$. Gaussian decay, then exponential time decay, then $v \le 2/9$ prove termination. These are the accepted tail and Cauchy arguments with each true exponential separately replaced by its upper bound E.

There are $P = 4JN$ potential pairs. A nonzero-character pair may be omitted only if its entire two-power contribution has bound

$$b = 4hn^a (m + h)^2 E(\delta n^2 (m - h)) \le \eta/(8P).$$

Add every omitted bound to B_{discard}, giving $B_{\text{discard}} \le \eta/8$. Evaluate all other integrated Taylor polynomials by the stated exact coefficient recurrence using outward rectangles. Refine both interval precision and all primitive series remainders until the final rectangle's sum of coordinate halfwidths satisfies $R_{\text{arith}} \le \eta/4$. For this fixed finite expression all input and primitive intervals shrink; continuity and outward arithmetic imply $R_{\text{arith}} \to 0$. Precision doubling therefore terminates, and the sum of the five raw components is at most $3\eta/4$. Choose an integer output scale B such that $5\, 2^{-B} \le \eta/8$ and round each component upward by $\text{up}_B(x) = 2^{-B} \lceil 2^B x \rceil$. The reported radius is their literal exact sum. It obeys

$$|y - \Lambda_q(s)| \le \sum_{x \in \{R_{\text{arith}}, B_n, B_t, B_Q, B_{\text{discard}}\}} \text{up}_B(x) \le \frac{3\eta}{4} + 5\, 2^{-B} \le \frac{7\eta}{8} < \eta.$$

The output retains exact rational midpoints, rounded components, B, raw interval radius, parameters, input metadata and source hashes. The raw analytic bounds are exactly reconstructible; they need not be serialized as potentially enormous fractions. The output precision varies with the request and introduces no fixed reporting floor. □

This fixed-conductor sample result uses no hypothesis on zeros. It does not remove a downstream fixed-node Fourier alias floor, supply an all-conductor theorem or a positive arithmetic determinant, or prove GRH.

2 Adaptive sampling parameters and precision termination

The primitive theta/Mellin identity with root number one yields

$$\Lambda_q(s) = \sum_{n\geq 1} \chi_q(n) n^a \int_1^\infty e^{-\pi n^2 t/q} \big(t^{(s+a)/2-1} + t^{(1-s+a)/2-1}\big)\, dt.$$

On $D = [-1,2] + i[-4/3, 4/3]$, both exponents p satisfy $|\Re p| \leq 2$ and $|\Im p| \leq 2/3$. For exact $x \geq 0$, put $E(x) = 2^{-\lfloor x \rfloor}$. The elementary inequality $e > 2$ gives $e^{-x} \leq e^{-\lfloor x \rfloor} \leq E(x)$ and $E(x) \leq 2e^{-x\log 2}$. Use this solely as a pointwise nonincreasing upper bound; it is not multiplicative and has no asserted exponential integral identity.

Integration by parts gives $\int_1^\infty t^2 e^{-xt}\, dt = e^{-x}Q(x)$ for $Q(x) = x^{-1} + 2x^{-2} + 2x^{-3}$. For $a = 0$ the bound $n^a e^{-\delta n^2/2} \leq C_0 = 1$ is immediate. For $a = 1$, $n \leq (1+n^2)/2$ and $n^2 e^{-\delta n^2/2} \leq 2/(e\delta) \leq 1/\delta$ give $C_1 = (1+1/\delta)/2$. For $n = n_0 + j$, the inequality $n^2 \geq n_0^2 + 2n_0 j$ therefore bounds the omitted-n sum by

$$2C_a Q(\delta n_0^2) \frac{e^{-\delta n_0^2/2}}{1 - e^{-\delta n_0}} \leq 2C_a Q(\delta n_0^2) \frac{E(\delta n_0^2/2)}{1 - E(\delta n_0)} = B_n.$$

Take $n_0 = N + 1 \geq 2$; then $\delta n_0 \geq 6/5$ and the latter denominator is positive. Substituting $t = Tu$ in each retained-n time tail gives

$$B_t = \sum_{n=1}^{N} 2n^a T^3 E(\delta n^2 T) Q(\delta n^2 T).$$

The two tails are disjoint because the omitted-n tail uses the whole integral and the time tail uses only retained n.

For $j = 0, \ldots, J-1$ and $\ell = 0, \ldots, 3$ take

$$m = 2^j \left(1 + \frac{2\ell+1}{8}\right), \qquad h = \frac{2^j}{8}, \qquad T = 2^J.$$

These panels cover $[1, T]$ and have $h/m \leq 1/9$. On $|z - m| = \rho = m/2$, principal log is analytic, $\Re z \geq m/2$, and $|\arg z| \leq 1$. For $f(z) = e^{-\pi n^2 z/q} z^p$, the angular multiplier is at most $e^{2/3} \leq 3$, whence

$$|f(z)| \leq 3B(m) E(\delta n^2 m/2), \qquad B(m) = \max\{(3m/2)^2, (m/2)^{-2}, 1\}.$$

Cauchy's coefficient estimate, cancellation of odd terms, and geometric summation yield the B_Q in Theorem 1. Including zero-character or later discarded terms in that sum is harmless. Gaussian, time, and geometric decay prove termination of the sequential first-success searches with $B_n, B_t, B_Q \leq \eta/8$. For fixed q and $L = \max\{1, \log(1/\eta)\}$ their asymptotic scales can be chosen as $N = O(\sqrt{L})$, $T = O(L)$, $J = O(\log L)$, and $K = O(L)$. Polynomial prefactors in N, T, J contribute

only $O(\log L)$ to the logarithm of the error estimates; this is a truncation-size statement, not a rational-arithmetic bit-complexity bound.

For real $t \in [m-h, m+h]$, $t \geq 1$ and $t^{\Re p} \leq (m+h)^2$. The bound for a whole pair, including both powers, is therefore $4hn^a(m+h)^2E(\delta n^2(m-h))$. There are $P = 4JN$ potential pairs; summing only bounds at most $\eta/(8P)$ gives $B_{\text{discard}} \leq \eta/8$. No partial recurrence or isolated exponential is discarded. For active pairs retain the coefficient recurrence

$$r_{-1} = 0, \quad r_0 = 1, \quad r_{j+1} = \frac{(h/m)(p - \lambda m - j)r_j - \lambda m(h/m)^2 r_{j-1}}{j+1}, \qquad \lambda = \pi n^2/q.$$

Coefficient comparison in $tf' = (-\lambda t + p)f$ proves that the integrated Taylor polynomial is

$$2he^{p\log m - \lambda m} \sum_{j=0}^{K} \frac{r_{2j}}{2j+1}.$$

Outward arithmetic encloses its exact value, including the signed character weight and both powers.

Use an isolated p-bit dyadic context, with all primitive remainders refined as p increases. Put $\delta_p = 2^{-p-16}$. Each arctangent series separately stops when its next-term bound is at most δ_p, giving total weighted analytic tail width at most $20\delta_p$ in Machin's formula, before outward dyadic rounding. Each reduced logarithm series separately has tail at most δ_p; for $x = 2^k v$, the combined bound is $(1 + |k|)\delta_p$, before rounding. For the complex exponential, after reduction to $A \leq 1/2$, choose the Taylor order so the bound

$$\frac{A^{H+1}}{(H+1)!}\frac{1}{1 - A/(H+2)}.$$

is at most δ_p. Choose H by finite exact search; its denominator is positive. Add a symmetric exponential error enclosure to each coordinate before outward squaring. The fixed guard16 is not itself an acceptance bound: the final complete rectangle radius must satisfy $R_{\text{arith}} \leq \eta/4$. All interval caches must include p and preserve exact input-type guards. The successful p and retry precisions/radii are recorded. Individual series lengths are determined by these, the exact inputs and the pinned source and are reconstructible by replay; they are not explicitly stored.

For fixed N,J,K and exact target point, the expression has finitely many operations and only nonzero fixed integer denominators. Each primitive input is bounded for sufficiently large p; exponential halving counts are consequently bounded. Exact primitive tails and dyadic rounding widths tend to zero. Induction through the finite continuous expression, including finitely many outward squarings, gives $R_{\text{arith}} \to 0$. Doubling p after a failed comparison terminates. The argument also covers true roots and nodes regenerated at every p, since their widths tend to zero. A fixed nondegenerate s rectangle has no corresponding shrinking-input claim.

For serialization, initialize

$$B = \max\{32, \text{bitlen}(\text{den}\,\eta) - \text{bitlen}(\text{num}\,\eta) + 8\}$$

and increase B until $5\,2^{-B} \leq \eta/8$. Write $\widehat{x} = 2^{-B}\lceil 2^B x \rceil$ for each of the five raw components. Their exact reported sum is at most $3\eta/4 + 5\,2^{-B} \leq 7\eta/8$, and every component remains an upper bound. The output records B, the raw interval radius and complete rectangle, N/J/K, discard threshold/pair counts, exact target metadata and source hashes. These reconstruct all raw analytic bounds and pair sets without serializing huge exact tail denominators. For the practical requests below B is87 and114, respectively. Exact endpoints remain rational; the mathematical specification

assumes an integer representation sufficient for the requested size. Dynamic rounding removes the old fixed reporting floor. For rational s, validate its exact coordinates in closed D before conversion: an outward dyadic enclosure of $\Im s = 4/3$ extends beyond D, but the analytic bounds concern the true point. Circle nodes satisfy the strict exact bounds $-7/10 \le \Re s \le 17/10$, $|\Im s| \le 6/5$.

The exact parameter-design checker gives

η	q	N	J	T	K	active pairs
10^{-24}	3	12	7	128	19	71
10^{-24}	4	14	8	256	20	65
10^{-24}	5	16	8	256	20	111
10^{-32}	3	14	7	128	25	84
10^{-32}	4	16	8	256	26	72
10^{-32}	5	18	8	256	26	129

with exact rational budgets and no special-function oracle. The arithmetic precision is selected from the actual interval radius, not inferred from this table. This extension retains the existing classical theta identity; it adds no zero-location, positive determinant, all-conductor, Lean or GRH claim. A downstream adaptive readout requires its own node-count, alias, degree, Fourier and optimization choices.